

COMPUTATIONAL FLUID DYNAMICS MODELLING OF VACUUM-ASSISTED RESIN INFUSION IN COMPOSITE SANDWICH PANELS DURING WIND TURBINE BLADE MANUFACTURING

Md Tusher Mollah^{1,*}, Maksim Larionov², Robert S. Pierce² and Jon Spangenberg¹

¹Department of Civil and Mechanical Engineering, Technical University of Denmark, 2800 Kgs. Lyngby, Denmark

²Department of Wind and Energy Systems, Technical University of Denmark, 4000 Roskilde, Denmark

*Email: mtumo@dtu.dk

Keywords: Vacuum-Assisted Resin Infusion (VARI), Composite Sandwich Panels, Computational Fluid Dynamics (CFD), Experimental Validation, Infusion Time, Flow Front

Abstract

The manufacturing of wind turbine blades predominantly features composite sandwich panels made through the Vacuum-Assisted Resin Infusion (VARI) process. These sandwich regions involve multiple components, which include a lightweight balsa or foam core with grooves and channels encased in non-crimp glass-fibre fabric skins. Understanding resin flow behaviour and infusion time through these materials is difficult as there are a large number of parameters to consider. Therefore, this study focuses on developing a Computational Fluid Dynamics (CFD) model to provide a more accurate prediction of resin flow for the investigation of a wide range of parametric changes. The CFD model is initially developed to simulate resin flow through the lower reinforcement layers and shallow grooves of composite sandwich panels. The simulated flow front has been compared with the experimental ones captured at the top and bottom of the fabric at different infusion times. The experimental and simulated results agree relatively well; providing a new venue for investigating resin infusion performance through complex features of the layup. Furthermore, the model has been exploited to investigate the influence of groove spacing on the flow front, resin-filled volume, and infusion time.

1. Introduction

Vacuum-Assisted Resin Infusion (VARI) is an essential process in the modern composite manufacturing industry, particularly in wind turbine blade manufacturing, which requires large-scale production of lightweight and efficient components. Understanding resin flow plays a crucial role in this process [1]. Significant areas of wind turbine blades comprise composite sandwich panels, which provide the twofold benefit of high strength and lightweight regions. The representative layup of these panels consists of two configurations of glass NCF, including unidirectional and biaxial layers, serving as composite skins [2]. These layers are positioned on both sides of a lightweight foam core with specific grooves and channels to improve formability and resin flow. Additional layers of resin distribution media, perforated release films, and peel plies are also used underneath the vacuum bags to support resin flow and demoulding during manufacture. Understanding the resin flow and filling of these sandwich regions is critical as it directly relates to the performance and quality of the wind turbine blade.

Computational Fluid Dynamics (CFD) models have shown considerable success in predicting resin flow for different infusion techniques, including VARI, resin injection pultrusion, resin film infusion, resin transfer moulding, etc. [3–8]. The majority of the CFD models employed for the modelling of resin transport in porous media rely on Darcy's Law, linking fluid velocity to pressure drop within the

medium, resulting in a single differential equation [9–11]. A detailed summary of the simulation studies within the scope of the VARI process has been outlined by Pierce [2]. Such modelling approaches are sufficiently helpful in predicting the resin flow and approximate infusion time within the porous media; however, they neglect some considerable features [12]. For example, the interplay of the groove channel flow with the fabric-reinforced layers or the through-thickness flow behavior is commonly overlooked.

In this study, a CFD model has been developed to simulate resin flow through shallow grooves and the lower reinforcement layer of non-crimp glass-fiber fabric skin. The simulated results are compared with experiments to analyze the flow fronts at the bottom and top of the reinforcement layer. Furthermore, the effect of the cut-to-cut distances of shallow grooves on the flow fronts, resin-filled volume, and infusion time has been investigated. Section 2 outlines the methodology of the study, providing details of the experiments conducted and the CFD model. Section 3 presents the results of the study. Finally, the conclusions drawn from the results are summarized in Section 4.

2. Methodologies

2.1. Single channel flow experiments

In order to validate the CFD models, a series of single channel flow experiments were conducted. Four layers of $250 \times 900 \text{ mm}^2$ biaxial glass fabric were sandwiched between an acrylic plate with a single machined channel and a tempered glass mold. The average thickness of the fiber preform was 2.2 mm. Permeability values for the fabric in three principal directions were determined from an independent set of experiments consistent with previous studies [13,14]: $K_1 = 1.89 \times 10^{-10} \text{ m}^2$; $K_2 = 6.14 \times 10^{-11} \text{ m}^2$; $K_3 = 1.093 \times 10^{-11} \text{ m}^2$; with a K_1 orientation of $45.19 \pm 1.38^\circ$ relative to the roll direction. The $900 \times 250 \times 14.7 \text{ mm}^3$ transparent polymethyl methacrylate (PMMA) plate was used instead of real core material for two reasons: firstly, to simplify the validation case for modelling (avoiding resin absorption into the core) and secondly, to allow for video observation from both sides of the experiments.

The machined channel was produced with dimensions $1 \times 3 \text{ mm}^2$ and a pressure sensor was placed as close to the channel inlet as possible as shown in Figure 1-a, to help establish the inlet pressure boundary conditions over time for modelling. The whole configuration was covered by a vacuum bag, representative of a typical VARI process, and a 200 mm peel-ply breaking zone was used to arrest the flow between the end of the channel and the vacuum outlet.

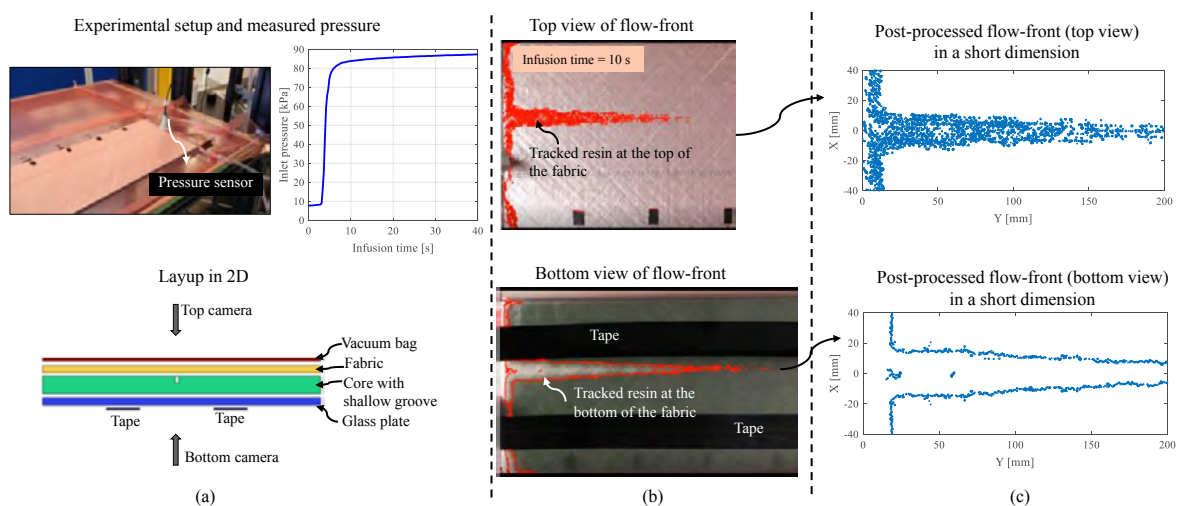


Figure 1. In-plane experiments and data post-processing. (a) Experimental setup, layup details, and the measured pressure profile over time; (b) Flow-front captured by the top and bottom camera after the tracking of the resin at the top and bottom of the fabric, respectively; (c) Post-processed flow-front in a short dimension to compare with the simulation results.

An 85% glycerol-water solution with density 1222 kgm^{-3} was used as the test fluid, with a viscosity of 0.0963 Pa.s at 20°C , which is representative of a typical low-viscosity resin for infusion. The pressure difference between the fluid reservoir at the inlet (open to ambient conditions) and the vacuum outlet (set with a constant 7500 Pa pressure) was used to drive the flow in the experiments. Videos recorded from both the top and bottom of the experiments were processed with a Python image-processing script based on the OpenCV library in order to track and visualize the flow front position over time as seen in Figure 1-b. The flow front propagation has been investigated for a reduced area of interest ($80 \times 200 \text{ mm}^2$), cf. Figure 1-c, in order to assist the CFD analysis described in the later section.

2.2. CFD model

The CFD model has been developed in FLOW-3D® [15] to simulate resin flow through the lower reinforcement layers and shallow grooves of composite sandwich panels. The model comprises a non-porous shallow groove in the core material and a porous channel at the top replicating the lower reinforcement layers of non-crimp glass fiber fabric skins, cf. Figure 2-a. The figure 2-a also shows the computational domain and implemented boundary conditions. The porous channel and the shallow groove are assigned pressure inlet boundary conditions at one end and pressure outlet at the other end. Time-dependent pressure boundary conditions have been implemented at the inlet according to the measured values in the experiment (cf. previous subsection) to better replicate experimental conditions. A constant pressure of 7500 Pa is applied at the outlet to mimic the vacuum outlet pressure during experiments. The top plane of the domain is a stationary wall representative of a rigid mold surface. The bottom plane (i.e., bottom of the porous channel and shallow groove) is also assigned to wall boundary conditions, reflective of the PMMA experiments. Although it should be acknowledged that this configuration neglects any fluid absorbance into the real core material. The rest of the planes are assigned to continuative boundary conditions having a zero velocity gradient while exiting the boundary.

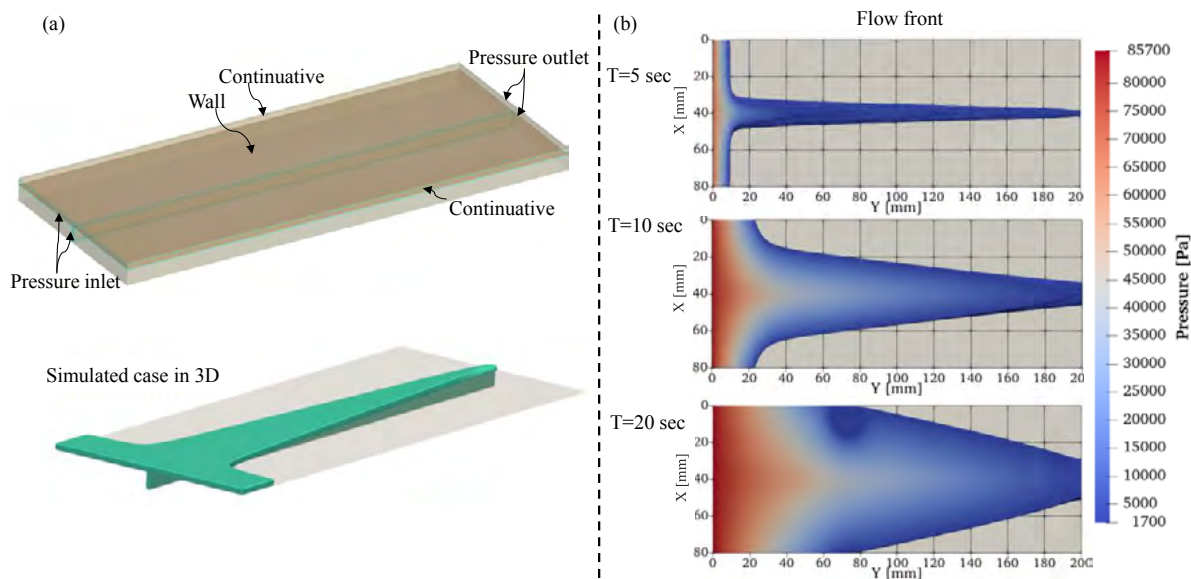


Figure 2. (a) CFD model geometry and the boundary conditions (top) and simulated sample in 3-dimension (bottom); (b) Flow front and pressure profile of the simulated cases at different infusion times.

The fluid flow is assumed to be transient, isothermal, and incompressible with a constant density ρ . The flow within the porous medium is solved based on the Darcian saturated drag model that adds a drag term to the momentum equations. Therefore, the flow dynamics are governed by the following mass and momentum conservation equations:

$$\nabla \cdot \mathbf{u} = 0 \quad (1)$$

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} = -\frac{1}{\rho} \nabla p + \mathbf{g} + \frac{\mu}{\rho} \nabla^2 \mathbf{u} - \mathbf{F} \quad (2)$$

where $\mathbf{u}$ is the velocity vector, p is the pressure, t is the time, $\mathbf{g}$ is the gravitational acceleration vector with components $(0, 0, -g)$; $g = 9.81 \text{ ms}^{-2}$ is the gravitational acceleration, and $\mathbf{F}$ is the resistance to flow in the porous medium, i.e., the drag term, and is given as follows:

$$\mathbf{F} = \mathbf{F}_d \mathbf{u} \quad (3)$$

where $\mathbf{F}_d$ is the drag coefficient vector of the porous media and is given as follows:

$$\mathbf{F}_d = \frac{\epsilon \mu}{\rho K} \quad (4)$$

where $\epsilon = (\epsilon_x, \epsilon_y, \epsilon_z)$ is the porosity vector equivalent to the volume fraction of the porous media, and K is the permeability of the porous media.

The permeability values found in experiments are different in x , y , and z directions (three principal directions) with a constant porosity ($0.506 = \text{volume fraction}$) in all directions, and they are implemented in the CFD model as directional porosity found by the equation of the drag coefficient, i.e., by Equation 4, where the permeability K is constant. Initially, the drag coefficient is calculated by $F_d = \epsilon \mu / \rho K$; where $\epsilon = 0.506$, and $K = 1.25\text{e-}10$ (derived from the sinusoidal transformation of the permeability along the principal direction, i.e., averaging K_1 and K_2 , as the angle of orientation is 45°). Finally, three porosity values are obtained by $\epsilon_i = F_d \rho K_i / \mu$; using the calculated F_d and the principal permeabilities (K_1, K_2, K_3). The porosity values corresponding to the permeability in principal directions are summarized in Table 1, which also includes specific details regarding the shallow cut and fabric dimensions.

The developed one fluid and void model uses the Volume of the Fluid method to track the free surface of the fluid through the low-pressure void domain, and the Finite Volume Method is used to discretize and solve the continuity and momentum equations. Figure 1-b showcases the simulated flow front for different infusion times, where the color represents the fluid pressure profile that changes over time.

Table 1. Properties of the fabric and details of the porous channel and shallow cut.

Case ID	Fabric thickness (mm)	Fabric volume fraction (-)	Fabric permeability (m^2)/porosity (-)			Dimension of the fabric or porous channel (mm^3)	Shallow cuts (-)	CtC spacing (mm)
			along x -direction	along y -direction	along z -direction			
1	2.22	0.506	1.89e-10/ 0.763	6.14e-11/ 0.248	1.093e-11/ 0.0668	80×200×2.22	1	N/A
2							2	10
3							2	20
4							2	40

3. Results and discussions

The flow fronts and the fluid filling between grooves over infusion time are investigated for two different configurations of the shallow grooves, i.e., single and double grooves, and are presented in Figures 3 to 7. Figures 3 to 5 present the flow fronts at different infusion times, specifically 5, 10, and 20 seconds, representing distinct stages of inlet pressure: sudden pressure buildup, the end of the buildup, and stabilized pressure, respectively. Figure 3 showcases experimental and simulated flow fronts at the bottom and top of the fabric at the infusion time of 5 seconds. The experiment shows that the flow fronts at the bottom and top of the fabric have significant differences, as expected, attributed to the more prominent flow along the groove and the slower flow along the z -direction within the fabric due to its lower permeability, cf. Figure 3-a. The simulated flow fronts at the bottom and top of the fabric

presented in Figure 3-b effectively capture the above phenomena. Figure 3-c provides the qualitative comparison between the experimental and simulated flow fronts. While simulations exhibit relatively good agreement, small discrepancies are observed. Specifically, the simulation slightly under-predicts the flow front length and width in both views; nevertheless, the resin flow within the fabric is more accurately predicted in the top view compared to the bottom view.

Figure 4 presents the flow fronts at the bottom and top of the fabric obtained from both experiment and simulation at an infusion time of 10 seconds, corresponding to the phase when the inlet pressure begins to stabilize. The experimental and simulated results, as illustrated in Figures 4-a and 4-b, respectively, exhibit similar patterns as observed in Figures 3-a and 3-b, apart from the fact that the flow front is more developed at this stage due to longer infusion time and the inlet pressure buildup. The qualitative comparison between the experimental and simulated flow fronts is presented in Figure 4-c. It is seen that the simulations tend to over-predict the flow compared to the experimental observations, regardless of the viewing perspective (i.e., top or bottom views).

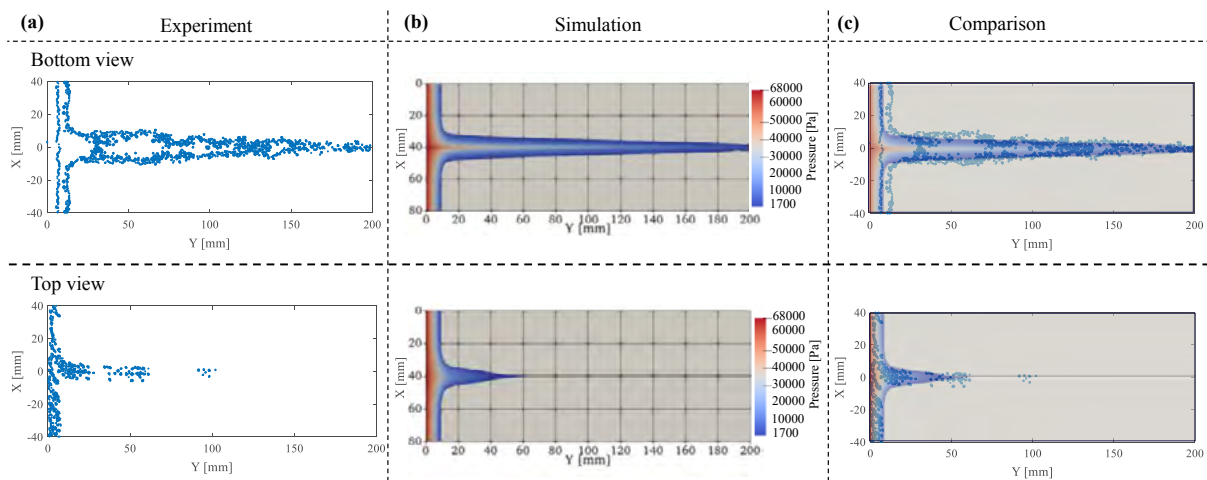


Figure 3. Flow fronts at the bottom and top of the fabric at an infusion time of 5 seconds. (a) Flow front captured during the experiment; (b) simulated flow fronts; and (c) qualitative comparison between experimental and simulated flow front.

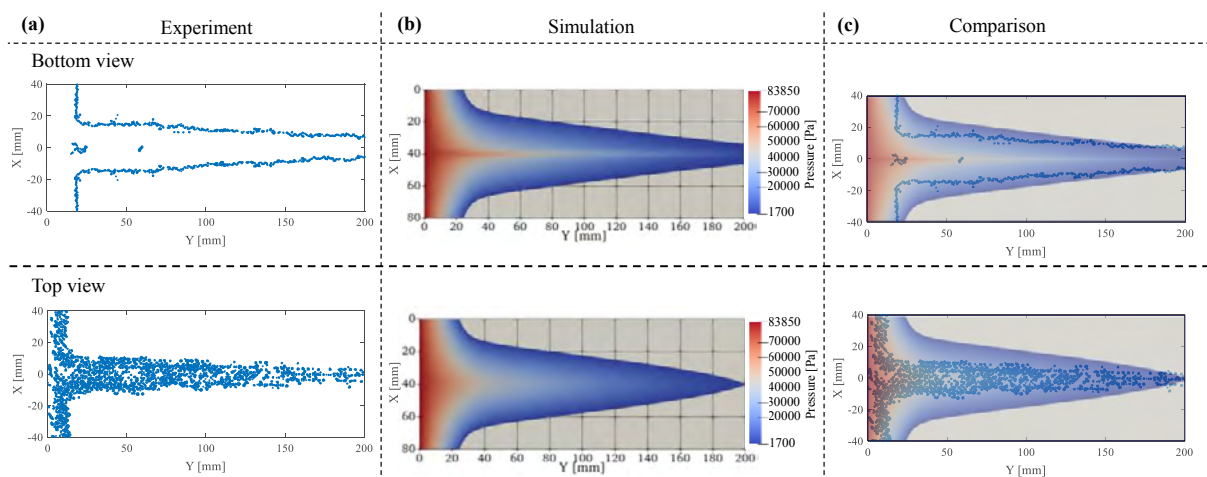


Figure 4. Flow fronts at the bottom and top of the fabric at an infusion time of 10 seconds. (a) Flow front captured during the experiment; (b) simulated flow fronts; and (c) qualitative comparison between experimental and simulated flow front.

Furthermore, the flow fronts are investigated at an infusion time of 20 seconds, corresponding to stable inlet pressure, and presented in Figure 5. Unlike previous observations, both experimental and simulated flow fronts at the bottom and top of the fabric are nearly identical, cf. Figures 5-a and 5-b. This similarity primarily arises from the dominance of flow through the fabric by high inlet- and groove-pressure, especially when looking into a smaller dimension. However, this phenomenon is required to be investigated to better understand this behaviour for different thicknesses, permeability of the fabric, and inlet pressure. The qualitative comparison between the experimental and simulated flow fronts in Figure 5-c gives similar information as seen in Figure 4-c that simulation tends to over-predict the flow in the fabric.

Overall, the simulations well predicted the nature of the flow fronts and their differences at both the top and bottom of the fabric. The simulations show the flow fronts and the resin flow in the fabric give the closest experiments when the infusion time is shorter. For larger times, the discrepancies become strong due to the dominating nature of the flow in the fabric from the channel in the investigated cases. These discrepancies can partially be due to issues of variability in permeability characterization and the difficulty in non-destructively assessing experimental fibre volume fractions. The predicted details are, however, sufficiently in agreement with the experiments, so the CFD model is used to investigate the flow fronts and filling between two grooves for different cut-to-cut (CtC) spacing of shallow grooves and are presented in Figures 6 and 7.

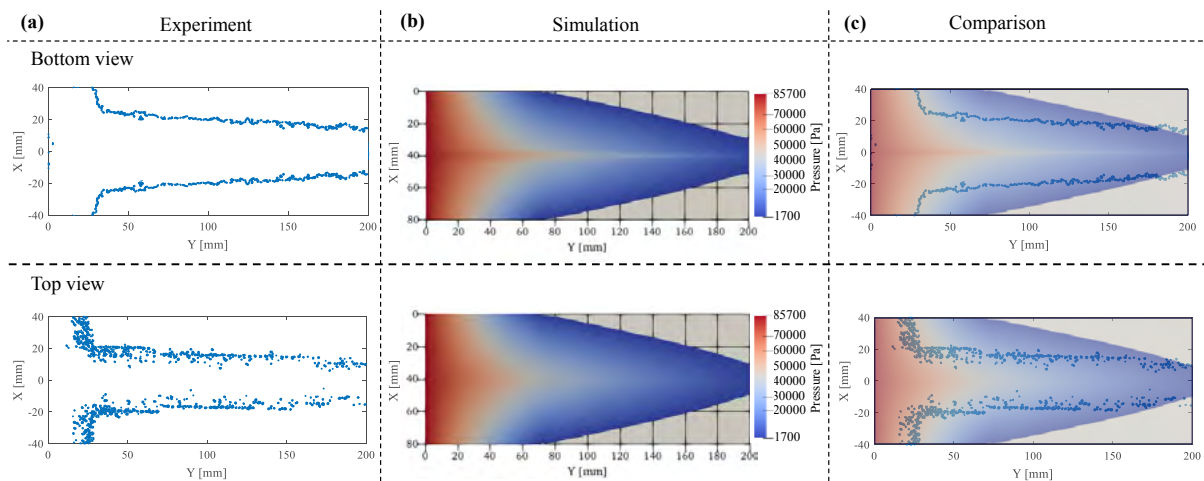


Figure 5. Flow fronts at the bottom and top of the fabric at an infusion time of 20 seconds. (a) Flow front captured during the experiment; (b) simulated flow fronts; and (c) qualitative comparison between experimental and simulated flow front.

Figure 6 illustrates the flow fronts at various infusion times for two grooves with different cut-to-cut (CtC) spacings: 10 mm, 20 mm, and 40 mm. The computational domain remains constant, as in the previous simulated case, at $80 \times 200 \times 2.22 \text{ mm}^3$, with the number of grooves limited to two. Realistically, a repeated pattern of grooves would be used in the true core materials. Therefore, while keeping computational complexity down, the presented results focus only on the filling between channels. Figure 6 shows that increasing CtC spacing increases the filling time between the channels, as expected. It can also be noticed that the minimum flow distance between channels (FD_{min}) changes for different CtC spacing, cf. Figure 6. Figure 7 quantifies the FD_{min} and presents it as a function of infusion time for both the top and bottom fabric surfaces. The FD_{min} reaches the length of the domain (200 mm) faster when the CtC is smaller, while it takes longer for the larger CtC spacing. Furthermore, it can be seen that the FD_{min} at the top and bottom of the fabric varies. For CtC 10 mm, the largest differences between the top and bottom FD_{min} are observed (blue curves), while the differences reduce when CtC spacing increases. For the largest investigated CtC 40 mm, the FD_{min} is identical at the top and bottom of the fabric throughout the filling (purple curves).

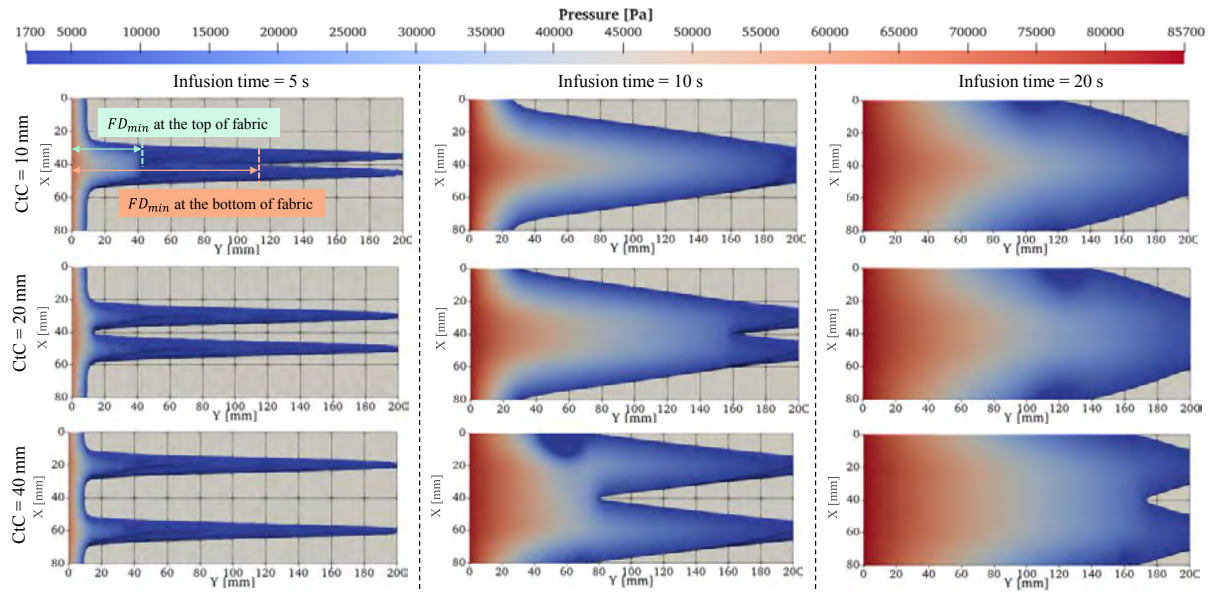


Figure 6. Flow fronts for different cut-to-cut (CtC) spacing of 10, 20, and 40 mm at different infusion times of 5, 10, and 20 seconds.

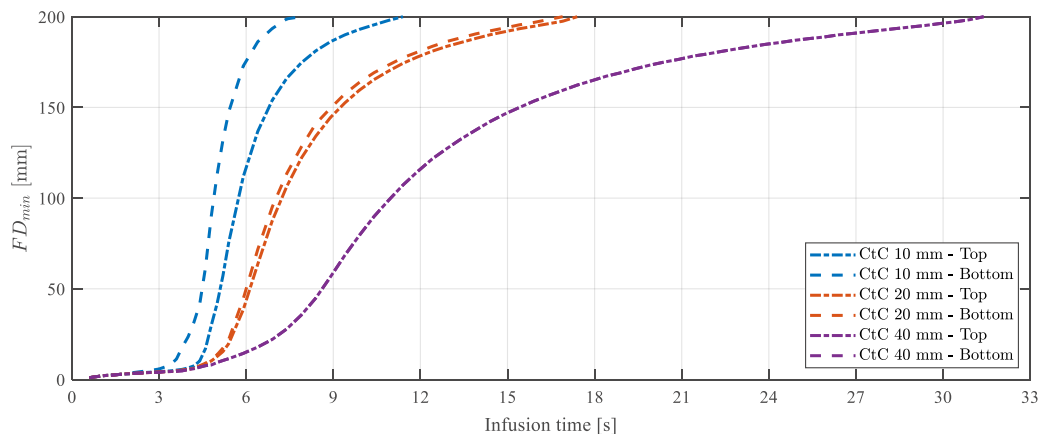


Figure 7. Minimum flow distance between channels (FD_{min}) at the top and bottom of the fabric as a function of infusion time for different cut-to-cut (CtC) spacing.

3. Conclusions

This paper investigates the Vacuum-Assisted Resin Infusion (VARI) process for manufacturing composite sandwich panels using experiments and simulations. The study focuses on the lower reinforcement layers and shallow grooves of the composite sandwich panels to analyze flow front profiles and infusion time. The simulated results accurately capture flow front profiles and patterns at both the bottom and top of the fabric, showing reasonable agreement with experimental data. Discrepancies are observed primarily for larger infusion times. Both experiments and simulations demonstrate that flow fronts at the bottom of the fabric are more pronounced than those at the top for smaller infusion times. However, as infusion time increases, flow fronts become more identical due to the dominance of flow through the fabric driven by high inlet and groove pressure. Furthermore, the model is utilized to simulate infusion with the presence of two shallow cuts, and the filling of fluid between two grooves is investigated for different cut-to-cut (CtC) spacings. As expected, the filling of fluid between two grooves takes less time for smaller CtC spacing. Additionally, it is noted that the minimum flow distance between the grooves varies at the top and bottom of the fabric. The differences are more pronounced when the CtC spacing is smaller, and they decrease when the CtC spacing is increased.

In future work, the simulated results can be compared with experiments involving various configurations of the channel, through-thickness experiments, and different fabric properties. This comparative analysis will provide deeper insights into the accuracy and applicability of the simulation model across diverse scenarios. Subsequently, a parametric study can be performed to better understand the process.

Acknowledgements

The authors would like to acknowledge the financial support of the Energy Technology Development and Demonstration Program (EUDP) for the “Opticore Platform” Udvikling Demonstration Project, Grant No. 64022-1041. The authors, MTM and JS, would like to thank FLOW-3D® for their support regarding licenses. Moreover, the author, MTM, would like to acknowledge Dr. Berin Šeta for the scientific discussion on CFD modelling.

References

- [1] Zhao C, Zhang G, Wu Y. Resin Flow Behavior Simulation of Grooved Foam Sandwich Composites with the Vacuum Assisted Resin Infusion (VARI) Molding Process. *Materials* 2012;5:1285–96. <https://doi.org/10.3390/ma5071285>.
- [2] Pierce RS. Challenges in permeability characterisation for modelling the manufacture of wind turbine blades. *IOP Conf Ser: Mater Sci Eng* 2023;1293:012009. <https://doi.org/10.1088/1757-899X/1293/1/012009>.
- [3] Sandberg M, Hattel JH, Spangenberg J. Flow-Induced Fibre Compaction in Resin-Injection Pultrusion. *Transp Porous Med* 2023;147:541–71. <https://doi.org/10.1007/s11242-023-01911-x>.
- [4] Spangenberg J, Larsen M, Rodríguez RR, Sánchez PMM, Rasmussen FS, Sonne MR, et al. THE EFFECT OF SATURATION ON RESIN FLOW IN INJECTION PULTRUSION: A PRELIMINARY NUMERICAL STUDY 2017.
- [5] Koefoed MS. Modeling and Simulation of the VARTM Process for Wind Turbine Blades n.d.
- [6] Han K, Jiang S, Zhang C, Wang B. Flow modeling and simulation of SCRIMP for composites manufacturing. *Composites Part A: Applied Science and Manufacturing* 2000;31:79–86. [https://doi.org/10.1016/S1359-835X\(99\)00053-6](https://doi.org/10.1016/S1359-835X(99)00053-6).
- [7] Celle P, Luca P, Drapier S, Bergheau J-M. Numerical modeling of infusion processes (LRI and RFI). vol. 52. 2007.
- [8] Bickerton S, Advani SG, Mohan RV, Shires DR. Experimental analysis and numerical modeling of flow channel effects in resin transfer molding. *Polym Compos* 2000;21:134–53. <https://doi.org/10.1002/pc.10172>.
- [9] Deléglise M, Binétruy C, Krawczak P. Solution to filling time prediction issues for constant pressure driven injection in RTM. *Composites Part A: Applied Science and Manufacturing* 2005;36:339–44. <https://doi.org/10.1016/j.compositesa.2004.07.001>.
- [10] Sandberg M, Yuksel O, Comminal RB, Sonne MR, Jabbari M, Larsen M, et al. 6 - Numerical modeling of the mechanics of pultrusion. In: Silberschmidt VV, editor. *Mechanics of Materials in Modern Manufacturing Methods and Processing Techniques*, Elsevier; 2020, p. 173–95. <https://doi.org/10.1016/B978-0-12-818232-1.00006-0>.
- [11] Sandberg M, Hattel JH, Spangenberg J. Simulation of liquid composite moulding using a finite volume scheme and the level-set method. *International Journal of Multiphase Flow* 2019;118:183–92. <https://doi.org/10.1016/j.ijmultiphaseflow.2019.06.014>.
- [12] Isoldi LA, Oliveira CP, Rocha LAO, Souza JA, Amico SC. Three-dimensional numerical modeling of RTM and LRTM processes. *J Braz Soc Mech Sci & Eng* 2012;34:105–11. <https://doi.org/10.1590/S1678-58782012000200001>.
- [13] Nedanov PB, Advani SG. A Method to Determine 3D Permeability of Fibrous Reinforcements. *Journal of Composite Materials* 2002;36:241–54. <https://doi.org/10.1177/0021998302036002462>.
- [14] Weitzenböck JR, Sheno RA, Wilson PA. Measurement of three-dimensional permeability. *Composites Part A: Applied Science and Manufacturing* 1998;29:159–69. [https://doi.org/10.1016/S1359-835X\(97\)00049-3](https://doi.org/10.1016/S1359-835X(97)00049-3).
- [15] FLOW-3D. FLOW-3D n.d. <https://www.flow3d.com/products/flow-3d/> (accessed May 7, 2024).